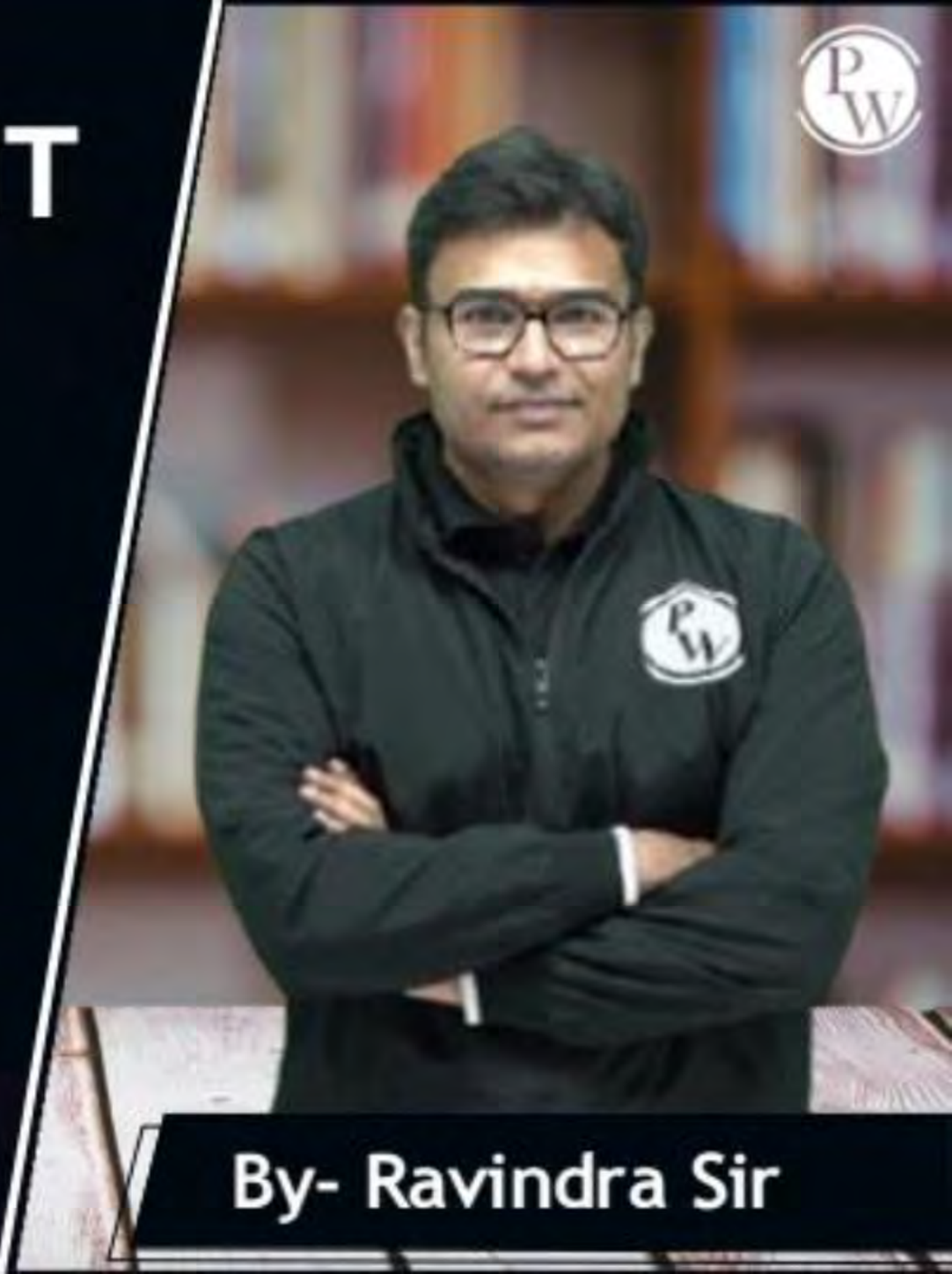
Computer Science & IT

COMPILER DESIGN
(CD)

SDT

Lecture No. 3



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

SOT

Topics to be Covered



- Topic
- Topic
- Topic

SDT

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

L-attributed definition:-

1) It allows both Synthesized as well as inherited attributes.
   But inherited attributes is restricted to inherit either from parent or from left sibling only

Ex: $A \rightarrow XYZ\ PQ$

$Z.i = f(A.i, X.i, Y.i)$ ✓

$Z.i = f(P.i, Q.i)$ ✗

2) Semantic actions can be placed anywhere on the RHS.

Ex: $A \rightarrow a\{\}$

$A \rightarrow B\{\}C$

$A \rightarrow \{\}CD$

They can be placed either in the beginning of in the middle of at the end.

3) Attributes are generally evaluated by traversing the parse tree depth first left to right.

Consider the following SDT :-

$A \rightarrow LM \{ L.i = f(A.i),\ M.i = f(L.S),\ A.S = f(M.S) \}$

$A \rightarrow QR \{ R.i = f(A.i),\ Q.i = f(R.S),\ A.S = f(Q.S) \}$

Not S

Not 'L'

SDT to store type information into symbol table

$D \rightarrow T\ L \quad \{L.in = T.type\}$

$T \rightarrow int \quad \{T.type = int\}$

$T \rightarrow char \quad \{T.type = char\}$

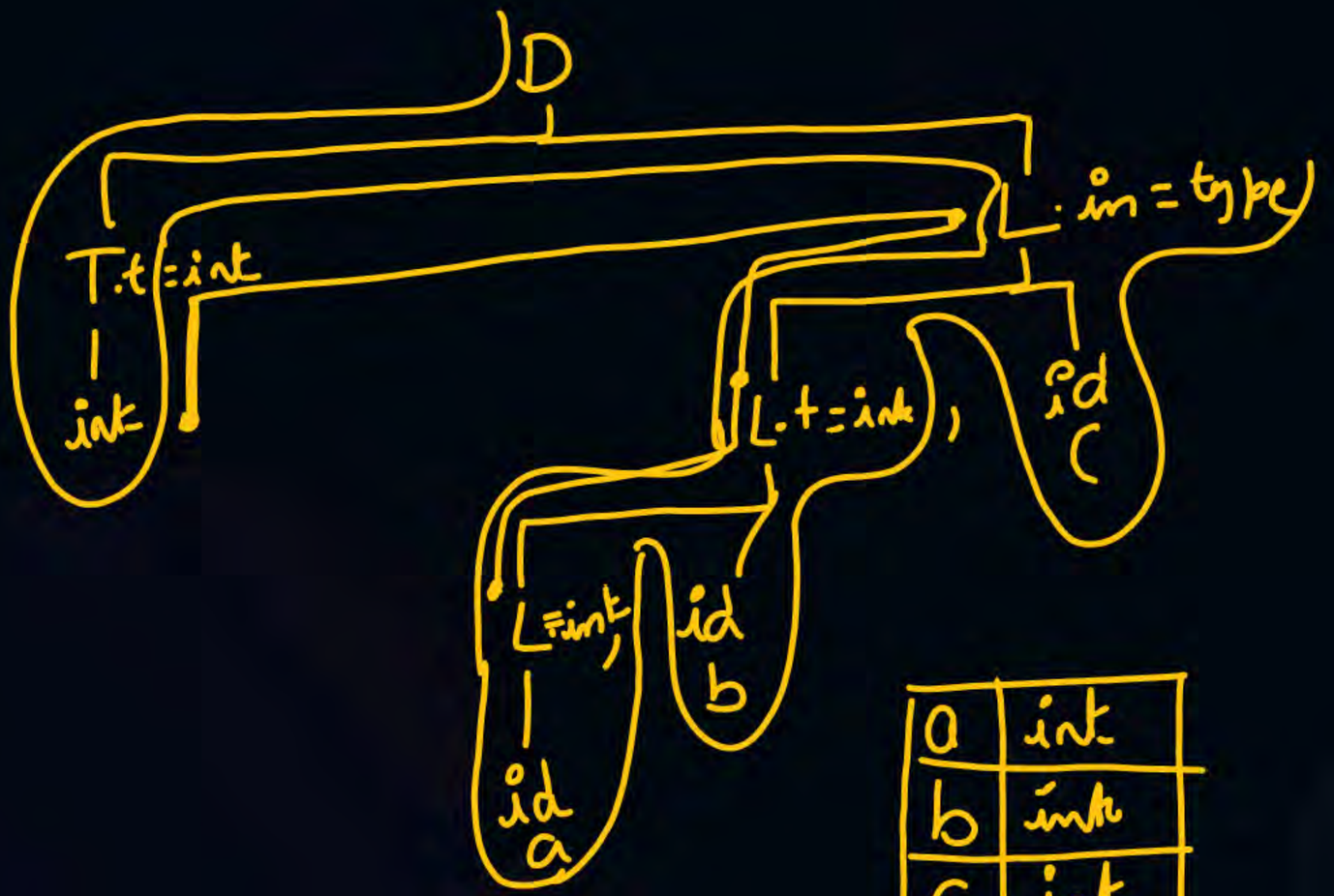
$L \rightarrow L_1, id \quad \{L_1.in = L.in,\ addtype(L_1.in, id)\}$

$L \rightarrow id \quad \{add(L.in, id.name)\}$

Evaluation of L attributed definition :-

1) Construct parse tree
2) Traverse the parse tree depth first left to right
3) Evaluate the inherited attributes when node is visited first time
4) Evaluate the Synthesized attributes if node is visited for the last time





int a, b, c

a	int
b	int
c	int

a	int
b	int
c	int

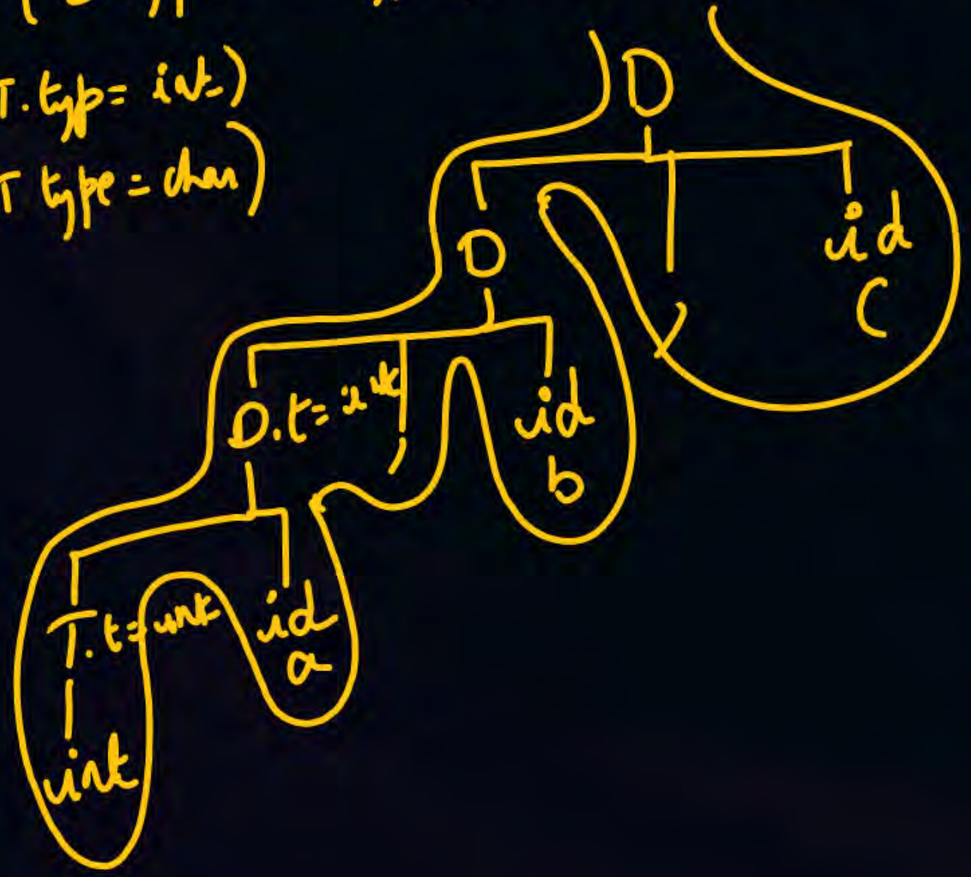
S attr def for storing the type info into Symbol tab

$D \rightarrow D_1$, id { D.type = D_1.type ; add_Type (D.type, id.name)

$D \rightarrow T$, id { D.type = T.type , add_type (T.type, id.name)

$T \rightarrow$ int { T.type = int }

$T \rightarrow$ char { T type = char }



int	a
int	b
int	c

THANK - YOU